University of Florida, Department of Mathematics

Probability PhD Exam, May 2008

1. Let $\mathscr{F}$ be a sigma field. Two random variables X and Y satisfy

$$\mathbb{E}[Y^2 \mid \mathscr{F}] = X^2, \qquad \mathbb{E}[Y \mid \mathscr{F}] = X.$$

Show that $X = Y$ a.s.

2. Let X and Y be two independent random variables with the same probability distribution $F(x)$. If $(X+Y)/\sqrt{2}$ has the same probability distribution $F(x)$, then

$$F(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\, \mathrm{d}y.$$

3. Let $f_n(x_1, \dots, x_n)$ be the probability density function of the random vector $(X_1, \dots, X_n)$ and $g_n(x_1, \dots, x_n)$ be the probability density function of the random vector $(Y_1, \dots, Y_n)$. We define

$$Z_n = \frac{g_n(X_1, \dots, X_n)}{f_n(X_1, \dots, X_n)},$$

if $f_n(X_1, \dots, X_n) > 0$, otherwise $Z_n = 0$. Show that $\{Z_n, n = 1, 2, \dots\}$ is a supermartingale.

4. Let $X_1, X_2, \dots$ be a sequence of i.i.d random variables with mean 0 and variance $\sigma^2 < \infty$. If $S_n = X_1 + X_2 + \cdots + X_n$, then

$$\lim_{n\to\infty} \mathbb{E}\left[\frac{|S_n|}{\sqrt{n}}\right] = \sqrt{\frac{2}{\pi}}\sigma.$$

5. Let B_t be a standard Brownian motion and define

$$D_n = \max_{n \le t \le n+1} |B_t - B_n|.$$

Prove that D_n/n converges to 0 with probability one.

6. Consider square integrable random vectors

$$Y = \big(y(1), y(2), \dots, y(d)\big)$$

and

$$Y_n = \big(y_n(1), y_n(2), \dots, y_n(d)\big), \qquad n \in \mathbb{N},$$

which satisfy

$$\lim_{n\to\infty} \mathbb{E}\Big(|y_n(j) - y(j)|^2\Big) = 0 \qquad \text{for } j = 1, 2, \dots, d.$$

Show that the mean of Y_n converges to the mean of Y, and the covariance matrix of Y_n converges to that of Y.

7. Suppose that $\{X_t\}$ and $\{Y_t\}$ are both standard Brownian motion and are independent. Let $\mathscr{F}_t$ denote the σ-field $\sigma(X_s, Y_s \mid s \in [0, t])$. Prove that

$$Z_t = X_t^2 Y_t - \int_0^t Y_u \, \mathrm{d}u$$

is an $\mathscr{F}_t$-martingale.

8. Let $\mathcal{F}$ be a sub-sigma field of $\mathcal{G}$ and Z be a nonnegative random variable defined on probability space $(\Omega, \mathcal{G}, \mathbb{P})$ with $\mathbb{E}_{\mathbb{P}}[Z] = 1$. Define

$$\mathbb{Q}(A) = \int_A Z \, \mathrm{d}\mathbb{P} \qquad \text{for } A \in \mathcal{G}.$$

a). Show that $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{G})$.

b). For a random variable X, we have

$$\mathbb{E}_{\mathbb{Q}}[X \mid \mathcal{F}] = \frac{\mathbb{E}_{\mathbb{P}}[ZX \mid \mathcal{F}]}{\mathbb{E}_{\mathbb{P}}[Z \mid \mathcal{F}]}.$$